Unemployment: Causes and Consequences

ECON 300 - Labour Economics

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Learning Objectives (1/2)

- Explain how imperfect information can generate the coexistence of unemployment and vacancies.
- Assess how technological change, especially the Internet, affects job search and matching.
- Show how demand shifts and sectoral change can generate unemployment.
- Summarize evidence on job displacement and its consequences.

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Learning Objectives (2/2)

- Describe models where firms pay above-market wages: efficiency wages, implicit contracts, and insider–outsider models.
- Explain the rationale for unemployment insurance/employment insurance and its effects on labour supply and unemployment.
- Summarize empirical evidence on UI/EI, incidence, and duration of unemployment.

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- How does imperfect information create frictional unemployment?
- Is online search improving matching efficiency?
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- Minimum wage laws
- Institutional factors such as unions or a large public sector
- Demand factors and seasonal unemployment
- Frictional unemployment arising from turnover and imperfect information
- Sectoral shifts causing structural unemployment
- Generous UI/EI affecting search incentives

Definitions

- **Frictional unemployment:** short-term unemployment arising from normal job search and turnover.
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Interpreting the Different Causes

- Not all unemployment has the same cause, so not all unemployment should be addressed with the same policy.
- Frictional unemployment may require better information and matching.
- Structural unemployment may require retraining or geographic mobility.
- Cyclical unemployment may require macroeconomic stimulus.

Real Example

A laid-off oil worker in Alberta may face structural unemployment if jobs are growing mainly in healthcare or technology rather than energy.

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Search Unemployment: Core Ideas

- Imperfect information delays matching between workers and firms.
- Searching for jobs has both costs and benefits.
- Workers adopt a **stopping rule** or **reservation criterion**: accept the first offer that meets a minimum acceptable standard.
- Optimal search balances expected marginal benefit and expected marginal cost.

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Why Search Takes Time

- Workers do not observe all vacancies instantly.
- Firms do not immediately identify the best applicants.
- Search time can improve job match quality, but it also delays income.

Interpretation

A longer search may be rational if it increases the chance of finding a better job. Unemployment is not always wasteful; some of it is part of efficient job matching.

Real Example

A recent graduate may reject a low-wage unrelated job and search longer for an entry-level job in their field because the long-run payoff is better.

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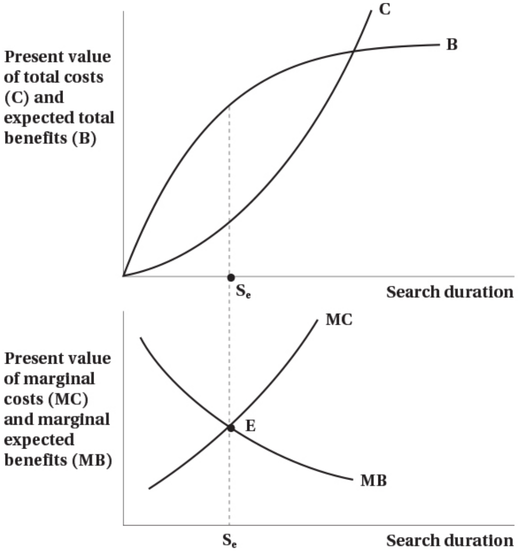
Figure 17.1: Optimal Job Search



Reading Figure 17.1

- In the top panel, total expected benefits of search rise at first but eventually flatten.
- Total costs rise as search continues.
- In the bottom panel, optimal search occurs where **marginal expected benefit equals marginal cost**.

Interpretation

The optimal search duration is not necessarily zero. It is efficient to keep searching until the extra expected gain from one more unit of search just equals its cost.

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Worked Example: Reservation Wage and Hazard

Julia faces three job offers with equal probability: \$2,400, \$2,200, and \$2,000 per month.

- Expected value of the next offer:

$$E[W] = \frac{2400 + 2200 + 2000}{3} = 2200$$

- Reservation wage:

$$W^R = 2200$$

- She rejects any offer below \$2,200 and accepts the first offer at or above \$2,200.
- Acceptance probability:

$$P(W \geq 2200) = \frac{2}{3}$$

Definition

The **hazard rate** is the probability that an unemployed worker exits unemployment in a given period, conditional on still being unemployed.

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- Higher search cost lowers the reservation wage.
- Better future offer prospects raise the reservation wage.
- Higher unemployment benefits may allow workers to search longer and hold out for better jobs.

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A worker with savings or EI benefits may reject a very low offer and continue searching, while a worker with no buffer may accept quickly.

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Imperfect Information: Market Implications

- Markets do not clear instantaneously.
- Wage dispersion can persist even with homogeneous workers and jobs.
- Individual firms may face upward-sloping labour supply because search frictions reduce worker mobility.

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Dynamic monopsony refers to a setting where individual firms have some wage-setting power because workers face frictions in moving between jobs.

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- Search frictions prevent immediate movement to the best-paying employer.
- This helps explain why similar workers may earn different wages in similar jobs.

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Two retail firms in the same city may pay different wages because workers do not instantly know all job offers and switching jobs is costly.

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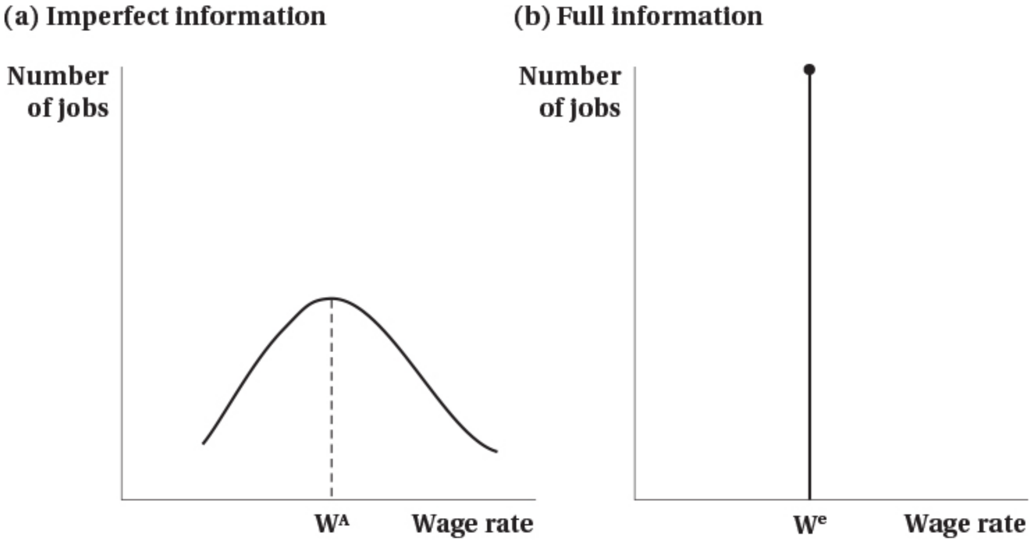
Figure 17.2: Wage Distributions Under Imperfect vs. Perfect Information



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- Under imperfect information, wages are spread over a distribution.
- Under full information, wages collapse to a single market-clearing wage.

Interpretation

Search frictions help explain both unemployment and persistent wage differences. Without frictions, workers would all move immediately to the highest available wage and wage dispersion would disappear.

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- Technology can raise contact rates between workers and firms.
- Better screening may improve match quality and shorten vacancy duration.

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Technology may reduce frictional unemployment by lowering search costs, but it can also raise application volume and screening complexity for employers.

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- But search is still imperfect:
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- Workers displaced from shrinking sectors do not automatically match into expanding ones.
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- Geography matters if jobs grow in different provinces or cities.

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A decline in manufacturing may reduce jobs in one region while healthcare and technology jobs expand elsewhere. Workers may need retraining or relocation to re-enter employment.

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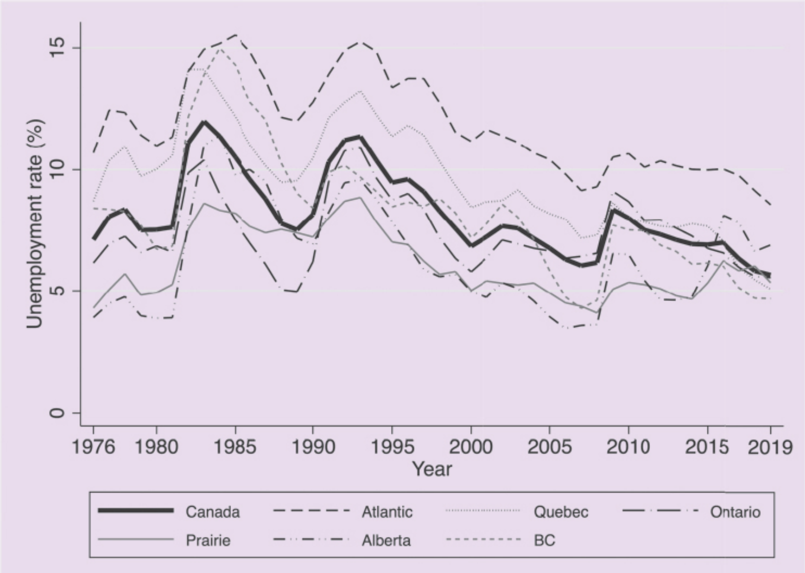
Figure 17.3: Regional Unemployment Rates, 1976–2019



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- Regional unemployment rates in Canada differ persistently.
- Some provinces face chronically higher unemployment than others.
- Regional shocks do not disappear instantly because moving is costly and local industry structures differ.

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Regional unemployment differences suggest that labour markets are not perfectly integrated. Geography is an important part of structural unemployment.

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- Re-employment frequently occurs at substantially lower wages.
- Losses can persist for many years.
- Older and less-educated workers often have lower re-employment probabilities.

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- Stress on family well-being and mental health

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Implicit Contracts: Risk Sharing

- Workers are often more risk-averse than firms because their human capital is not easily diversified.
- Firms may offer relatively stable wages across states of nature.
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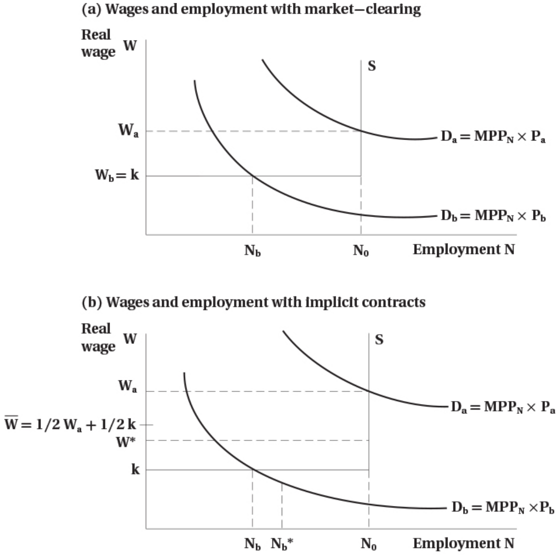
Figure 17.4: Implicit Contracts



Reading Figure 17.4

- In a market-clearing world, wages fluctuate with labour demand.
- Under implicit contracts, wages are smoother.
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Interpretation

Risk sharing can make workers better off ex ante, but it can also create layoffs and unemployment when demand is weak.

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Efficiency Wages: Core Mechanisms

- Higher wages may raise morale and effort.
- Higher wages may reduce shirking and turnover.
- Firms may use high wages to deter unionization or attract better applicants.
- In some models, better nutrition also raises productivity at low wage levels.

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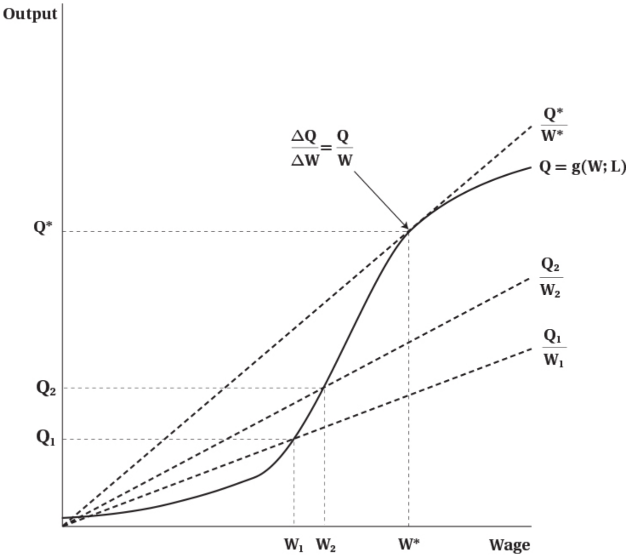
Figure 17.5: Determination of the Efficiency Wage



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- Output rises with the wage because worker efficiency increases.
- But the gain from raising the wage eventually diminishes.
- The optimal efficiency wage is where the gain from better performance is balanced against the higher wage cost.

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If firms pay efficiency wages, unemployment can persist because wages stay above the level that would clear the labour market.

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Insider–Outsider Theory

- Wages are bargained mainly with incumbent workers.
- Replacing insiders with outsiders is costly because of hiring, training, and disruption costs.
- Excess labour supply from outsiders may not push wages down much.

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Insiders are current employees with bargaining power. **Outsiders** are unemployed or non-employed workers who want jobs but have little influence on wage setting.

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Why Insider–Outsider Forces Matter

- Firms may prefer to keep current workers at high wages rather than replace them.
- This can generate wage rigidity and persistent unemployment.
- Outsiders may remain unemployed even if they are willing to work for less.

Real Example

A firm may keep experienced employees at negotiated wages because replacing them would require costly training and risk production disruptions.

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Economic Rationale for UI/EI

- UI/EI provides insurance against income loss from unemployment.
- It also affects incidence and duration of unemployment through incentives and liquidity.

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Channels: Incidence and Duration

- Higher benefits may increase separations by making unemployment less costly.
- For unemployed workers, higher benefits lower search costs and may lengthen optimal search.
- Empirically, exit rates often rise sharply as benefits approach exhaustion.

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UI/EI creates a trade-off: it provides valuable insurance but may also increase unemployment by affecting job separation and job search behaviour.

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Canada's UI/EI System: Key Features

- Established in 1940.
- Major reforms in 1971–72.
- Renamed Employment Insurance in 1996.
- Replacement rate around 55% for beneficiaries.
- Claimant types include frequent, long-tenured, and occasional users.

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The details of eligibility and benefit duration matter greatly for how EI affects both insurance value and labour market incentives.

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Behavioural Effects and Design Considerations

- Moral hazard vs. liquidity effects
- Repeat use and seasonal patterns
- Regional extended benefits
- Labour force participation effects

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- A well-designed system should protect workers from hardship.
- But it should also preserve incentives for efficient job search and employment.
- The optimal design balances social insurance against induced unemployment.

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If EI is too stingy, workers may be forced into poor matches quickly. If it is too generous for too long, some may delay search excessively. Good design tries to avoid both extremes.

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